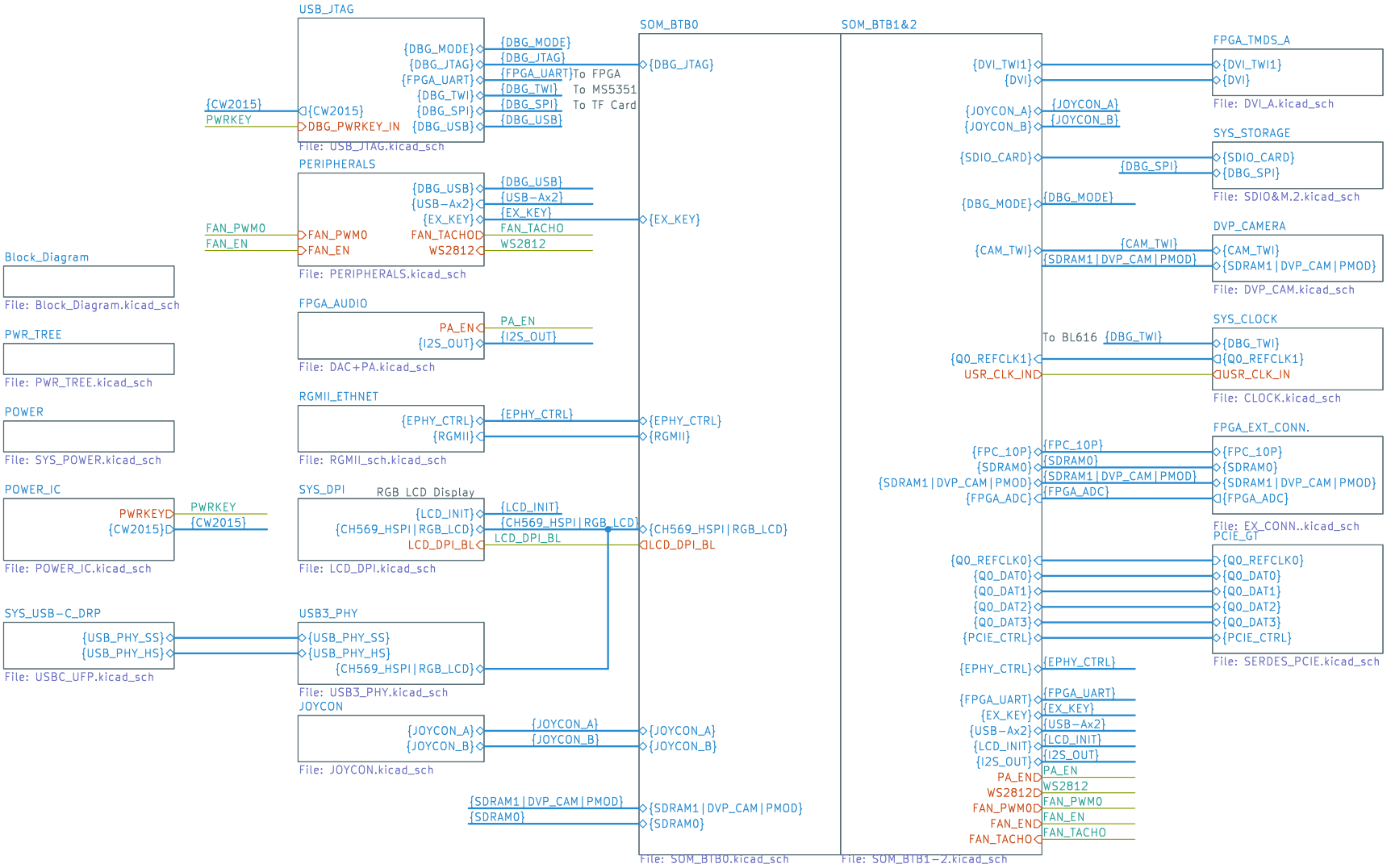
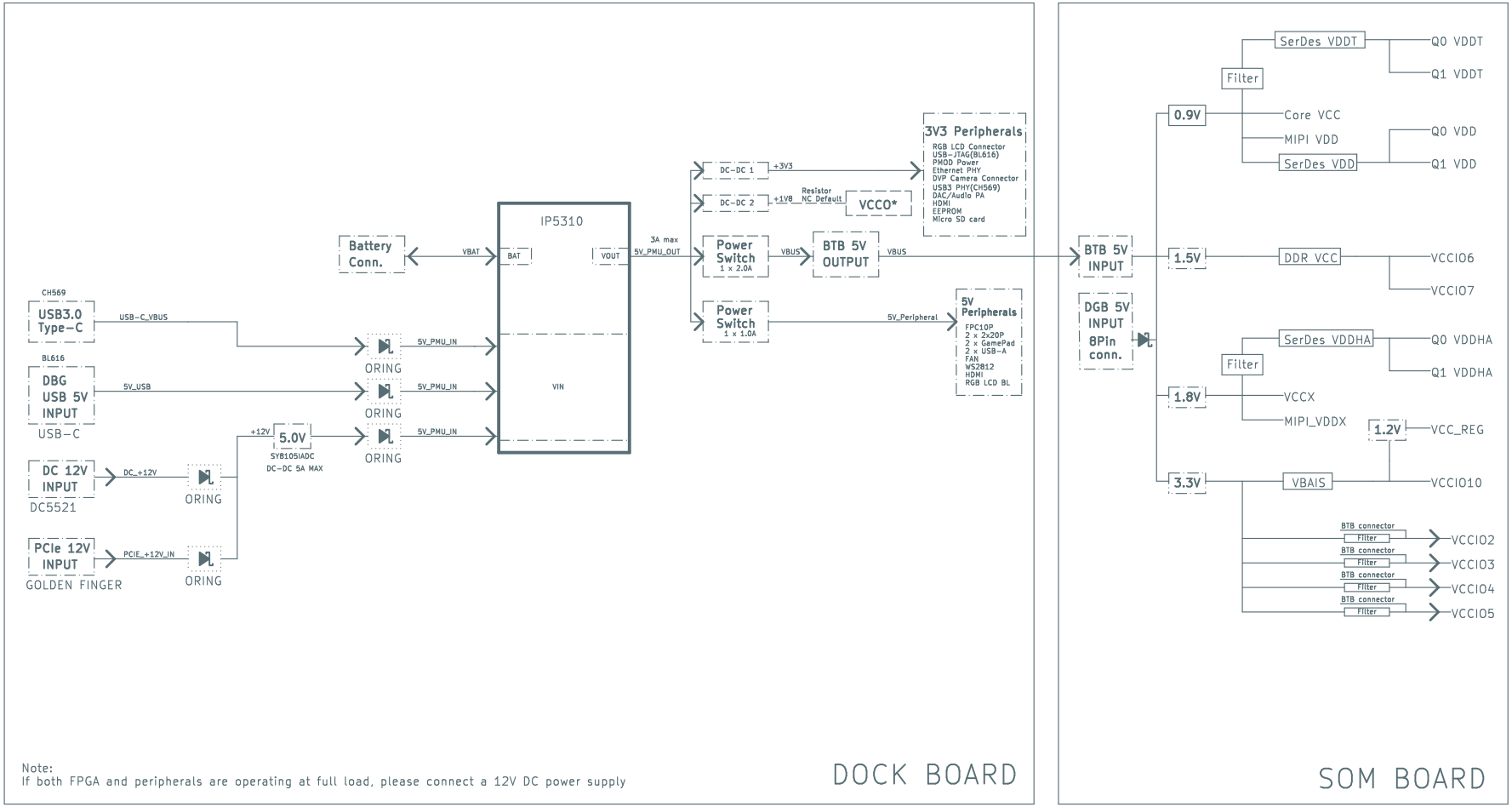


SiPEED

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Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 1/21

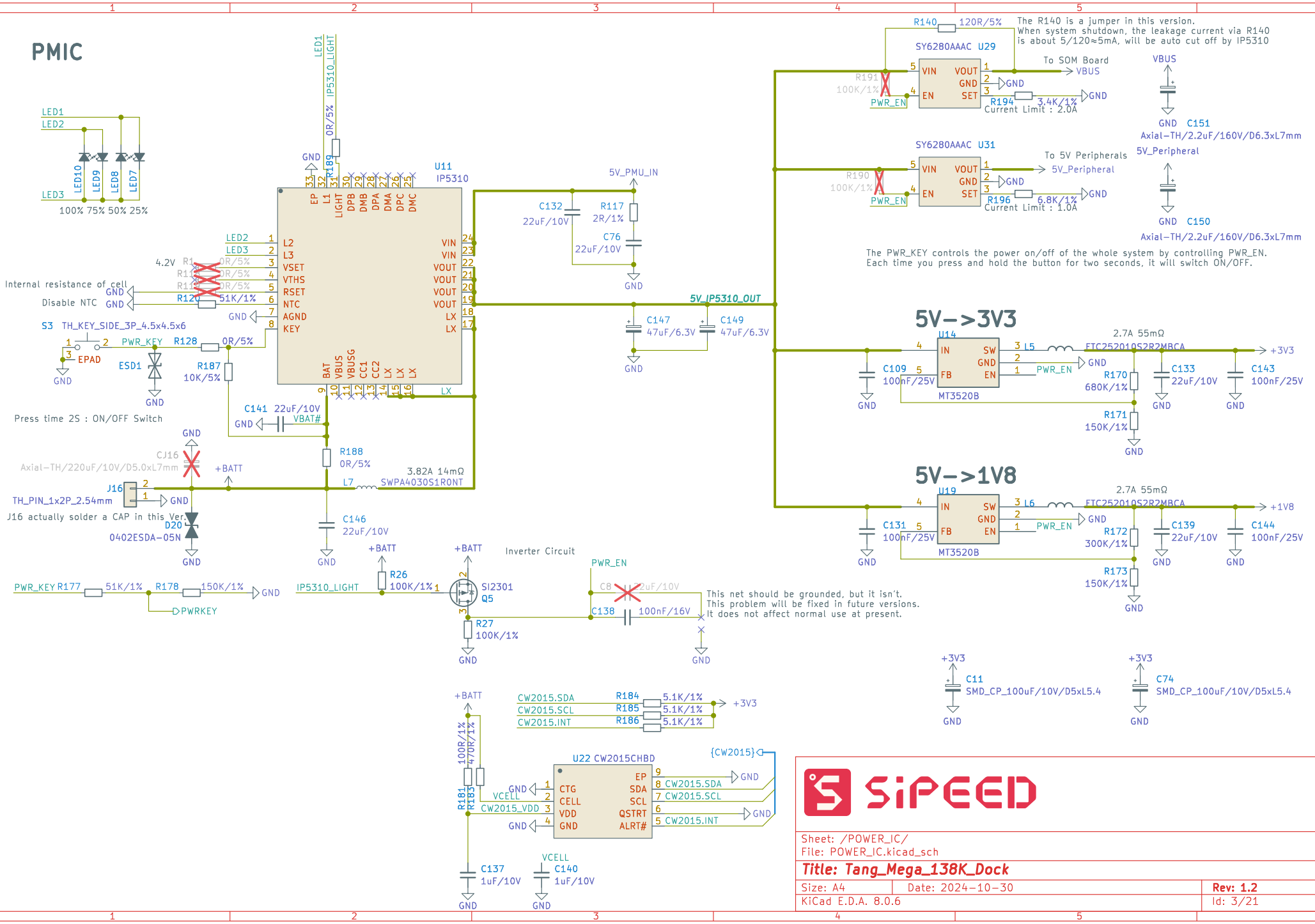


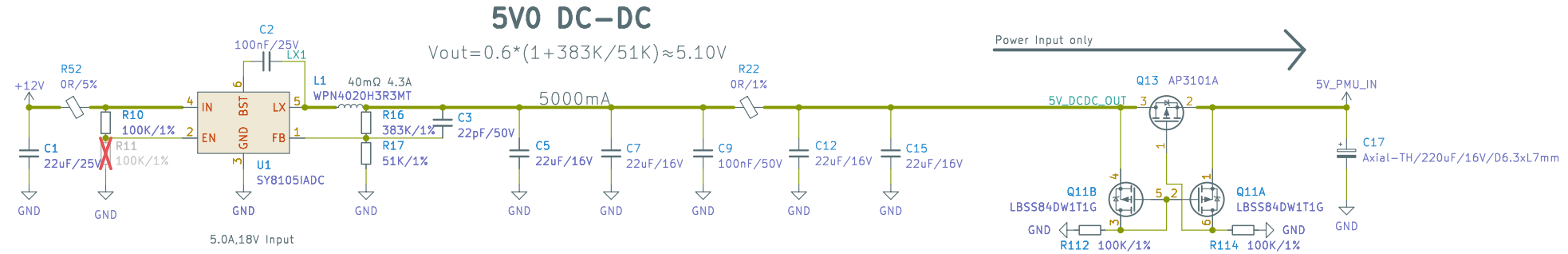
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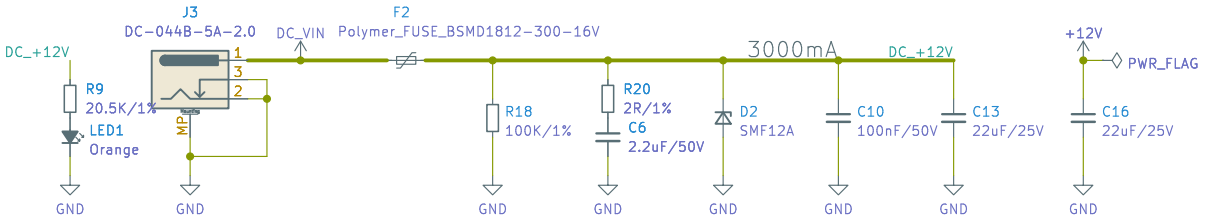
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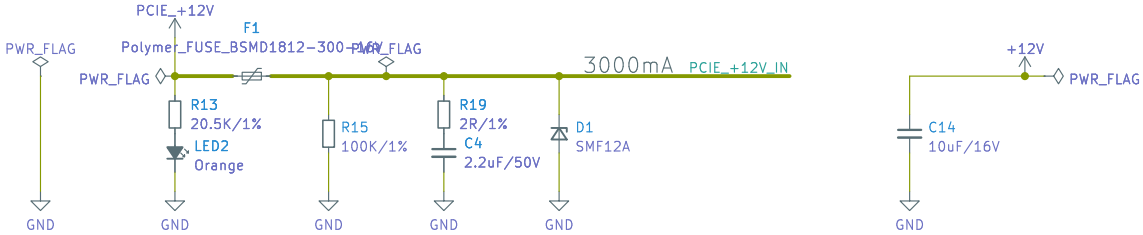
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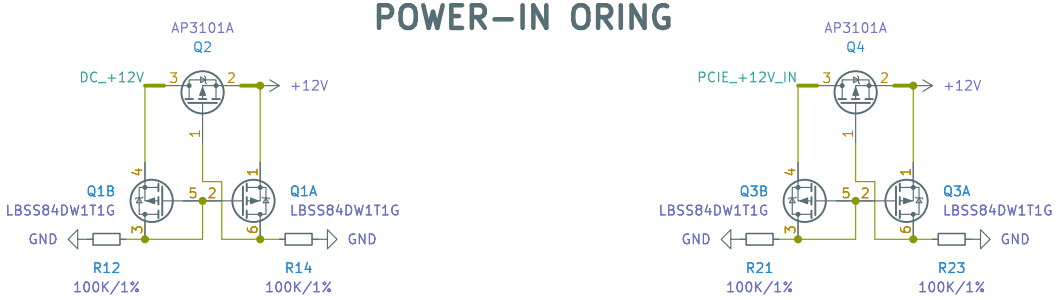
Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 2/21


PMIC
LED1
LED2
LED3
LED10
LED9
LED8
LED7
100% 75% 50% 25%
U11
IP5310
LED1
IP5310_LIGHT
0R/5%
R189
GND
EP L1 LIGHT DPB DMB DPA DMA DPC DMC
LED2 L2
LED3 L3
VSET
VTHS
RSET
NTC
AGND
KEY
VIN VIN VOUT VOUT VOUT VOUT LX LX
BAT VBUS VBUSG CC1 CC2 LX LX LX
4.2V R11 0R/5%
R12 51K/1%
Internal resistance of cell
Disable NTC
S3 TH_KEY_SIDE_3P_4.5x4.5x6
PWR_KEY
R128 0R/5%
EPAD
ESD1
R187 10K/5%
Press time 2S : ON/OFF Switch
C141 22uF/10V
VBAT#
R188 0R/5%
CJ16
Axial-TH/220uF/10V/D5.0xL7mm
+BATT
J16
TH_PIN_1x2P_2.54mm
J16 actually solder a CAP in this Ver.
D20
0402ESDA-05N
C146 22uF/10V
3.82A 14mΩ
L7
SWPA4030S1RONT
5V_PMU_IN
C132 22uF/10V
R117 2R/1%
C76 22uF/10V
5V_IP5310_OUT
C147 47uF/6.3V
C149 47uF/6.3V
R140 120R/5%
The R140 is a jumper in this version.
When system shutdown, the leakage current via R140
is about 5/120≈5mA, will be auto cut off by IP5310
SY6280AAAC U29
R191 100K/1%
PWR_EN
VIN VOUT GND SET EN
To SOM Board
VBUS
R194 3.4K/1%
Current Limit : 2.0A
VBUS
GND C151
Axial-TH/2.2uF/160V/D6.3xL7mm
SY6280AAAC U31
R190 100K/1%
To 5V Peripherals
5V_Peripheral
R196 6.8K/1%
Current Limit : 1.0A
GND C150
Axial-TH/2.2uF/160V/D6.3xL7mm
The PWR_KEY controls the power on/off of the whole system by controlling PWR_EN.
Each time you press and hold the button for two seconds, it will switch ON/OFF.
5V->3V3
U14
IN SW GND FB EN
L5
FTC252010S2R2MBCA
2.7A 55mΩ
+3V3
C109 100nF/25V
MT3520B
R170 680K/1%
R171 150K/1%
C133 22uF/10V
C143 100nF/25V
5V->1V8
U19
L6
C131 100nF/25V
R172 300K/1%
R173 150K/1%
C139 22uF/10V
C144 100nF/25V
+1V8
Inverter Circuit
PWR_KEY R177 51K/1%
R178 150K/1%
PWRKEY
IP5310_LIGHT
R26 100K/1%
SI2301
Q5
R27 100K/1%
C8 2uF/10V
C138 100nF/16V
This net should be grounded, but it isn't.
This problem will be fixed in future versions.
It does not affect normal use at present.
C11 SMD_CP_100uF/10V/D5xL5.4
C74 SMD_CP_100uF/10V/D5xL5.4
CW2015.SDA
CW2015.SCL
CW2015.INT
R184 5.1K/1%
R185 5.1K/1%
R186 5.1K/1%
100R/1%
47OR/1%
R181
R183
U22 CW2015CHBD
CTG CELL VDD GND EP SDA SCL QSTRT ALRT#
VCELL
CW2015_VDD
{CW2015}
C137 1uF/10V
C140 1uF/10V
SiPEED
Sheet: /POWER_IC/
File: POWER_IC.kicad_sch
Title: Tang_Mega_138K_Dock
Size: A4
Date: 2024-10-30
Rev: 1.2
KiCad E.D.A. 8.0.6
Id: 3/21









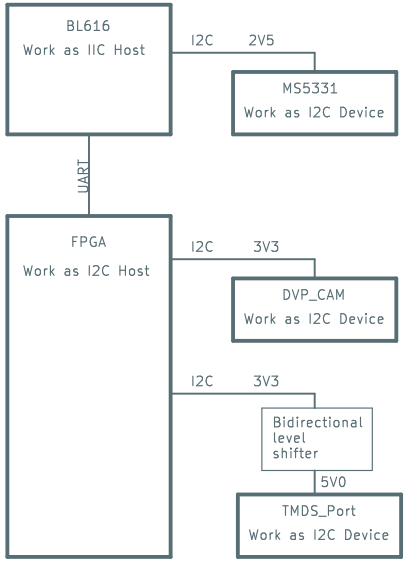
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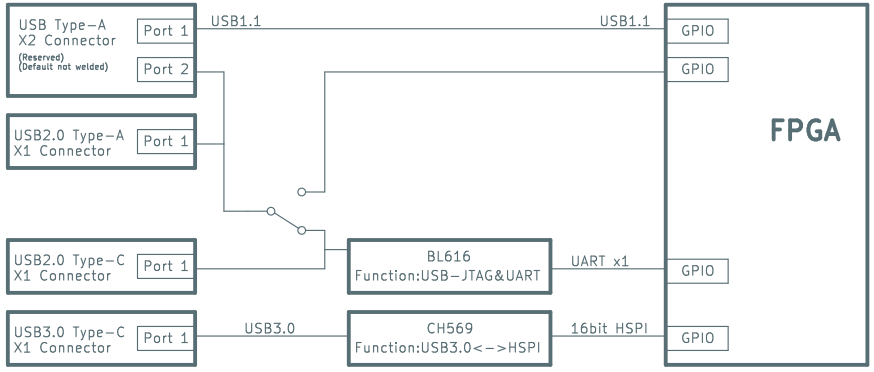
Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 4/21

IIC connection



USB connection

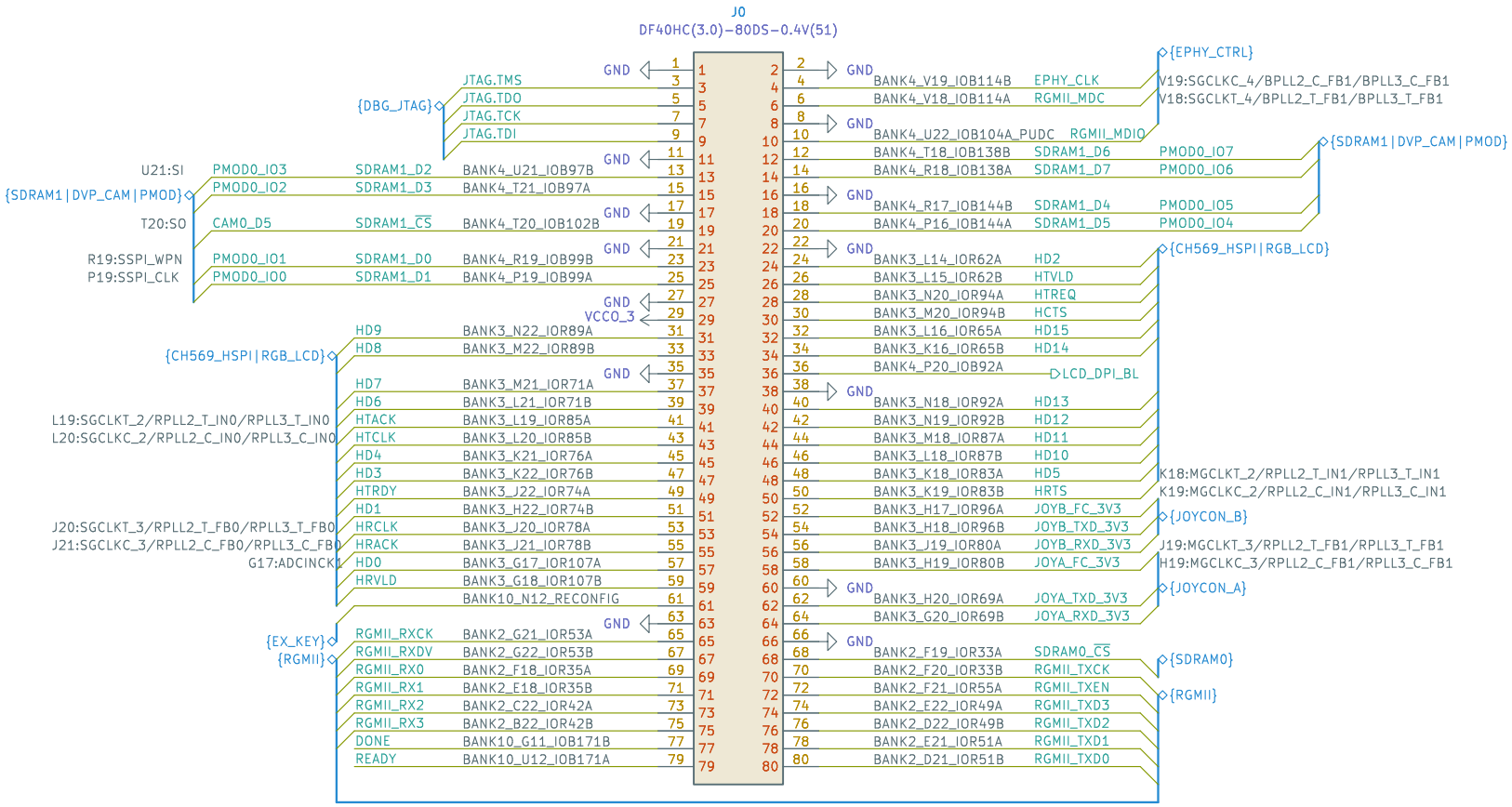


SiPEED

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Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 5/21

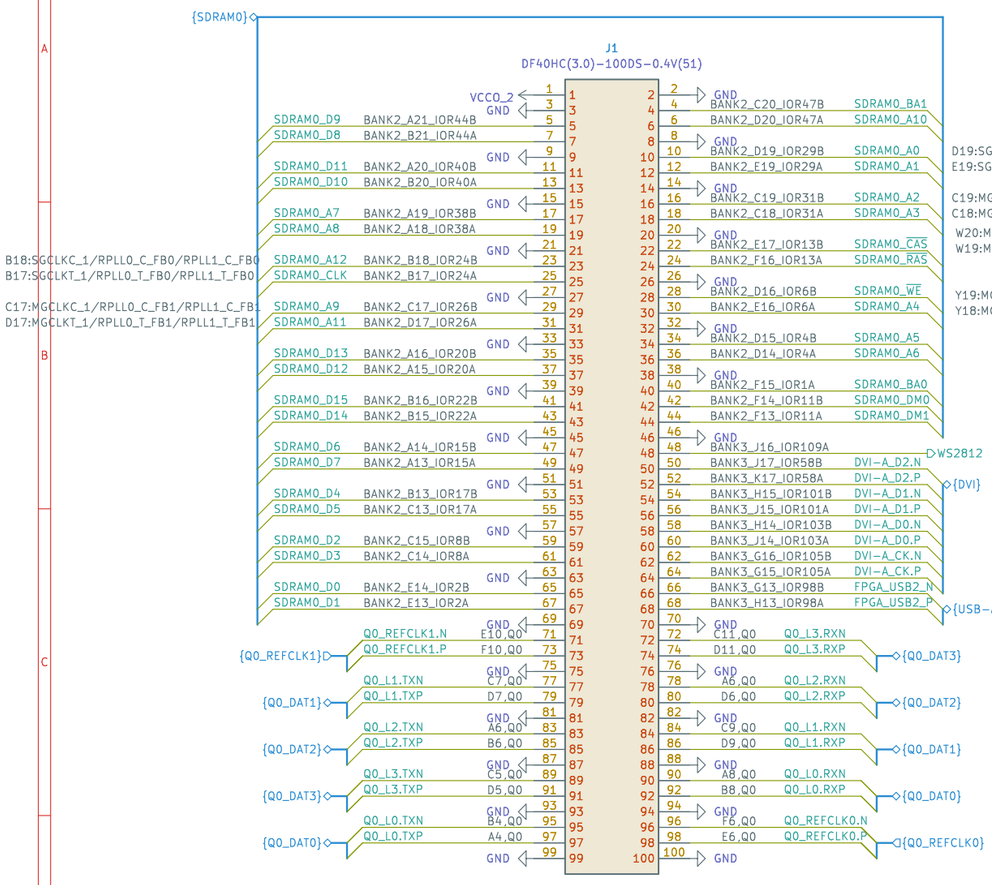


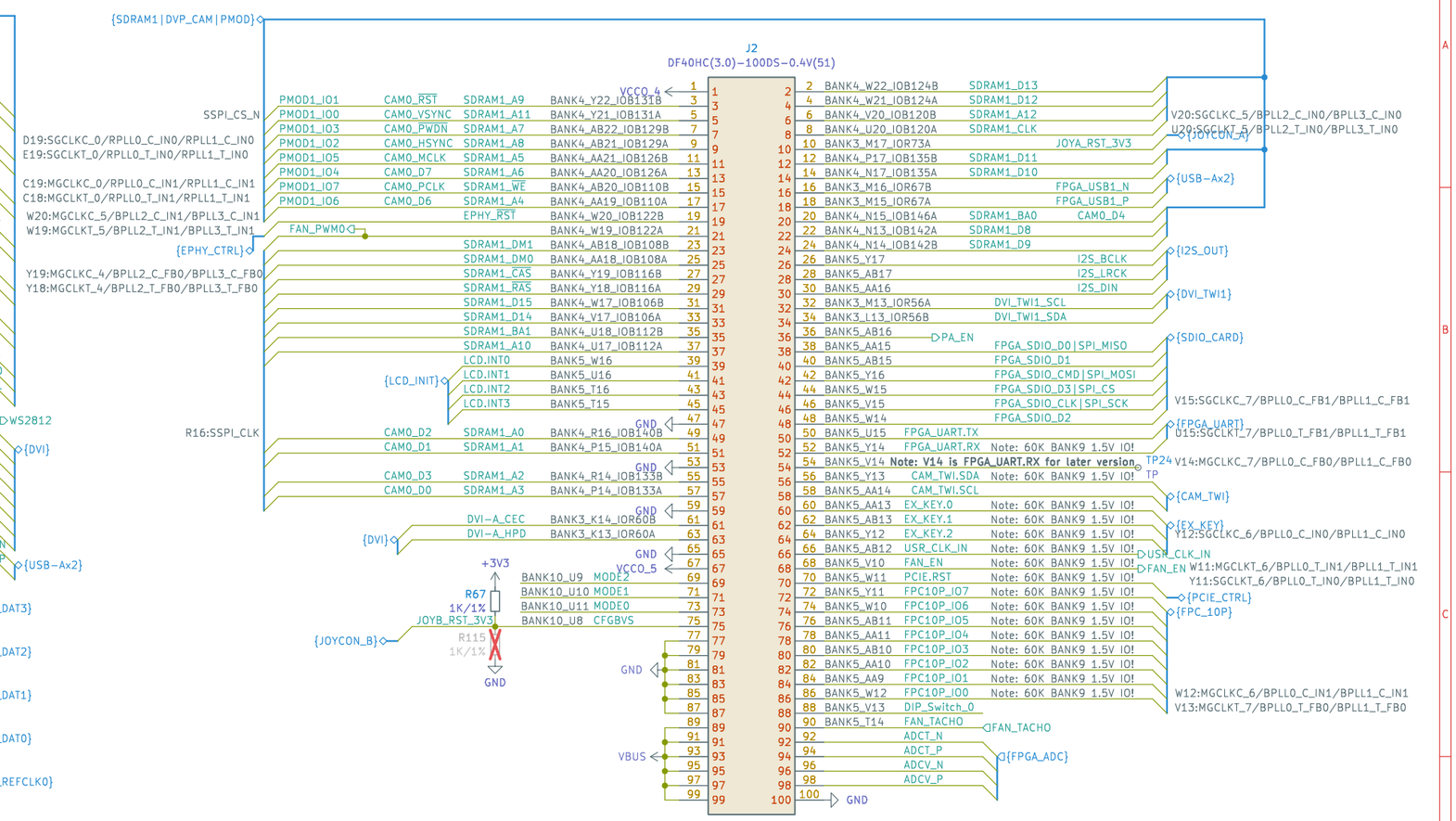
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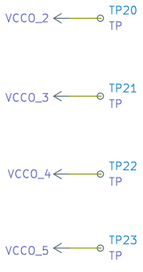
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Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 6/21







Switch

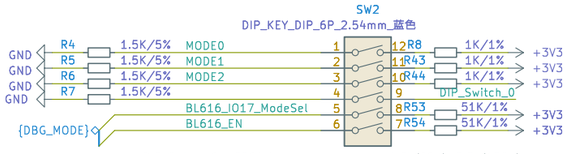


表 4-2 GW5AT-138/GW5AST-138 配置模式选择

配置模式	MODE[2:0][1]	Bus Width	相关说明
JTAG	XXX/101[3]	-	外部 Host 通过 JTAG 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置
MSPI[2]	001	x1,x2,x4	FPGA 作为 Master，通过 SPI 接口从外部 Flash（或其他器件）读取配置数据进行配置
SSPI[2]	010	x1,x4	外部 Host 通过 SPI 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置
Master SERIAL	000	x1	FPGA 作为 Slave 以前，通过 DIN 接口从外部读取配置数据进行配置
Slave SERIAL	111	x1	外部 Host 通过 DIN 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置
Master CPU	100	x8,x16,x32	FPGA 作为 Slave 以前，通过 DBUS 接口从外部读取配置数据进行配置
Slave CPU	110	x8,x16,x32	外部 Host 通过 DBUS 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置

Table 4-2 GW5AT-138/GW5AST-138 Configuration Modes

Configuration Mode	MODE[2:0][1]	Bus Width	Description
JTAG	XXX/101[3]	-	GW5AT & GW5AST series of FPGA products are configured by external Host via JTAG interface
MSPI[2]	001	x1, x2, x4	As a Master, FPGA reads data from external Flash (or other devices) via the SPI interface for configuration
SSPI[2]	010	x1, x4	GW5AT & GW5AST series of FPGA products are configured by external Host via SPI interface
Master SERIAL	000	x1	Before FPGA used as a Slave, FPGA reads data from external devices via the DIN interface for configuration
Slave SERIAL	111	x1	GW5AT & GW5AST series of FPGA products are configured by external Host via DIN interface
Master CPU	100	x8, x16, x32	Before FPGA used as a Slave, FPGA reads data from external devices via the DBUS interface for configuration
Slave CPU	110	x8, x16, x32	GW5AT & GW5AST series of FPGA products are configured by external Host via DBUS interface

BL616 SEL[0..1] SWITCH

NO.	CONFIG	SEL0(EN)	SEL1(MODE)
0	Disable BL616 Use other debugger	0	X
1	JTAG mode	1	1
2	Game mode	1	0



SiPEED

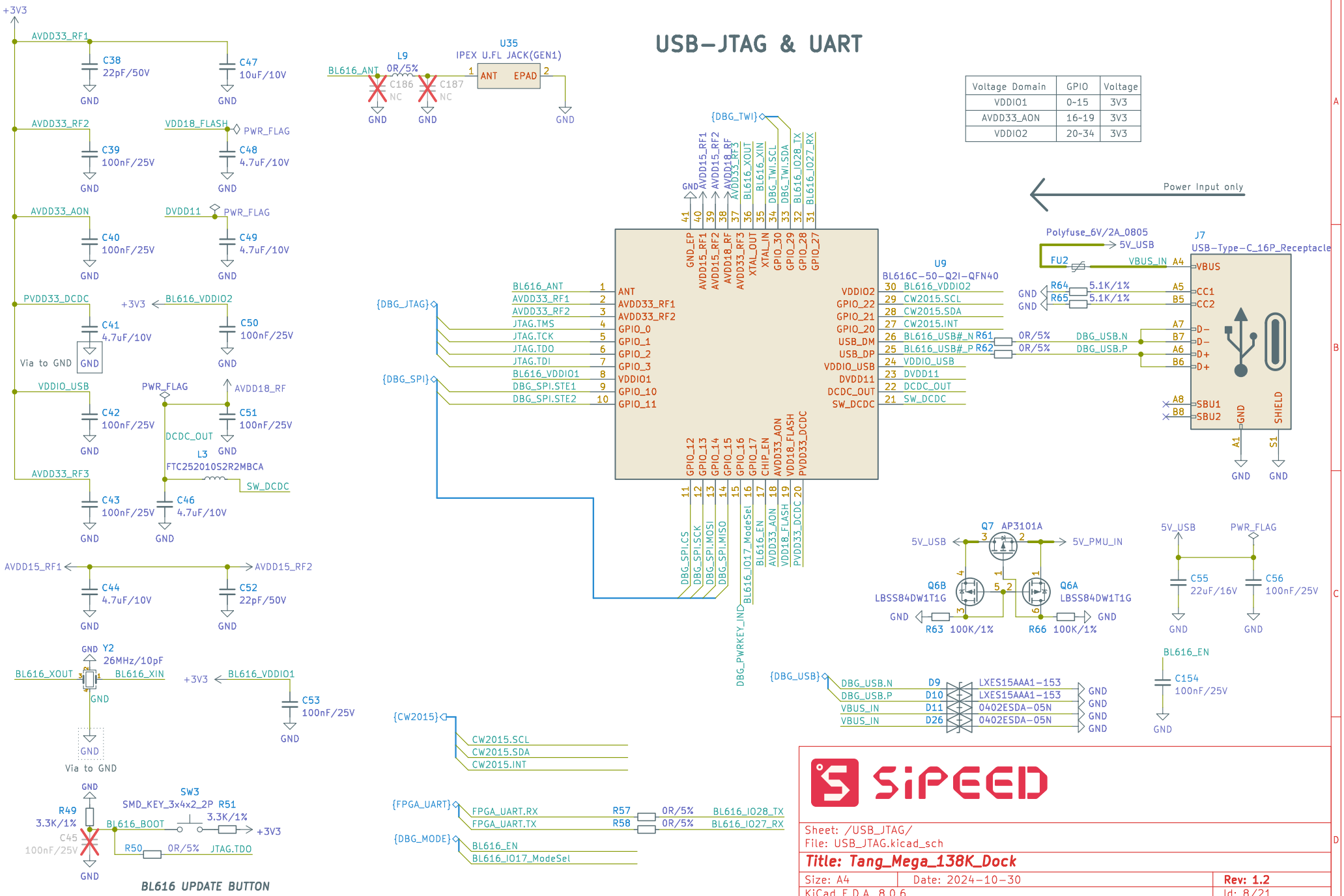
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Title: Tang_Mega_138K_Dock

Size: A3	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 7/21

USB-JTAG & UART

Voltage Domain	GPIO	Voltage
VDDIO1	0~15	3V3
AVDD33_AON	16~19	3V3
VDDIO2	20~34	3V3



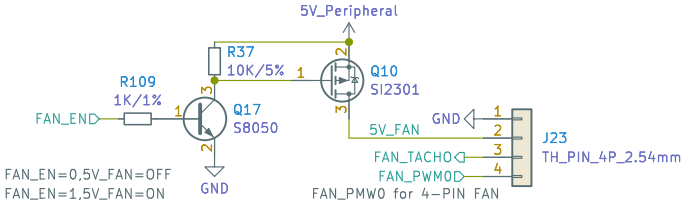
SiPEED

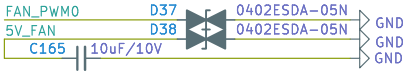
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Title: Tang_Mega_138K_Dock

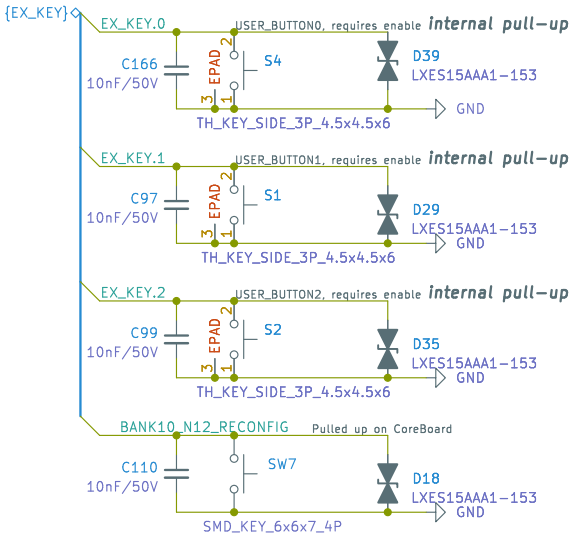
Size: A4	Date: 2024-10-30	Rev: 1.2
KiCad E.D.A. 8.0.6		Id: 8/21

FAN Connector

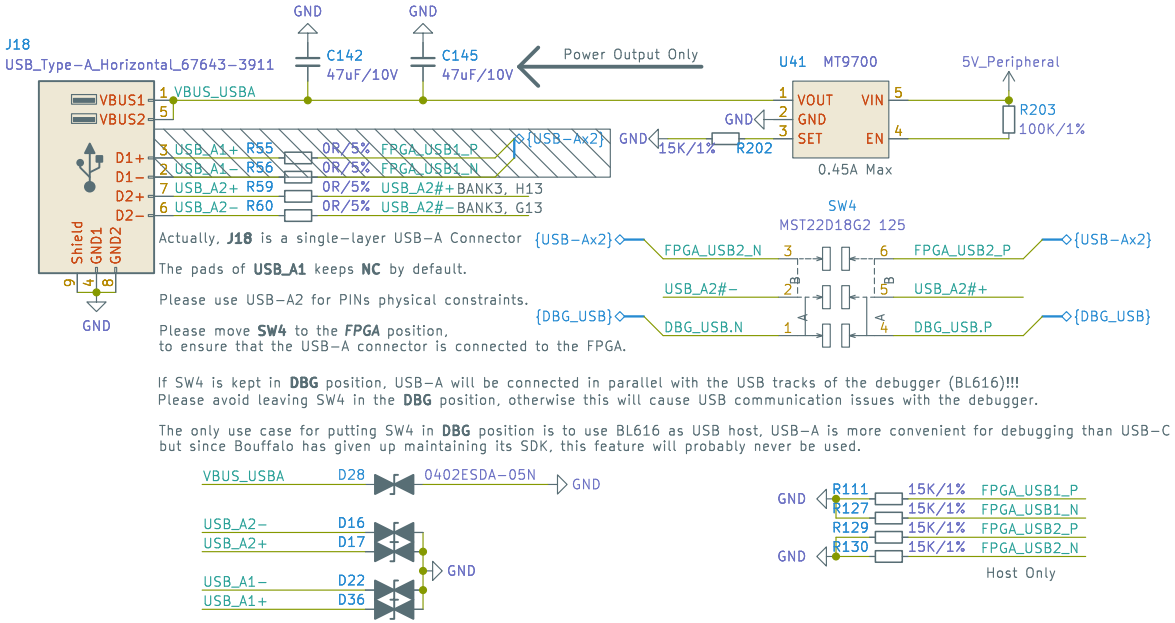




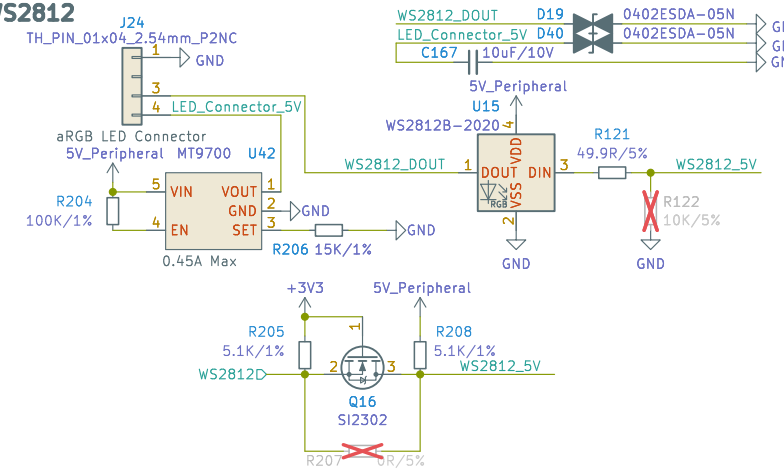
BUTTON



USB_A_x2



WS2812



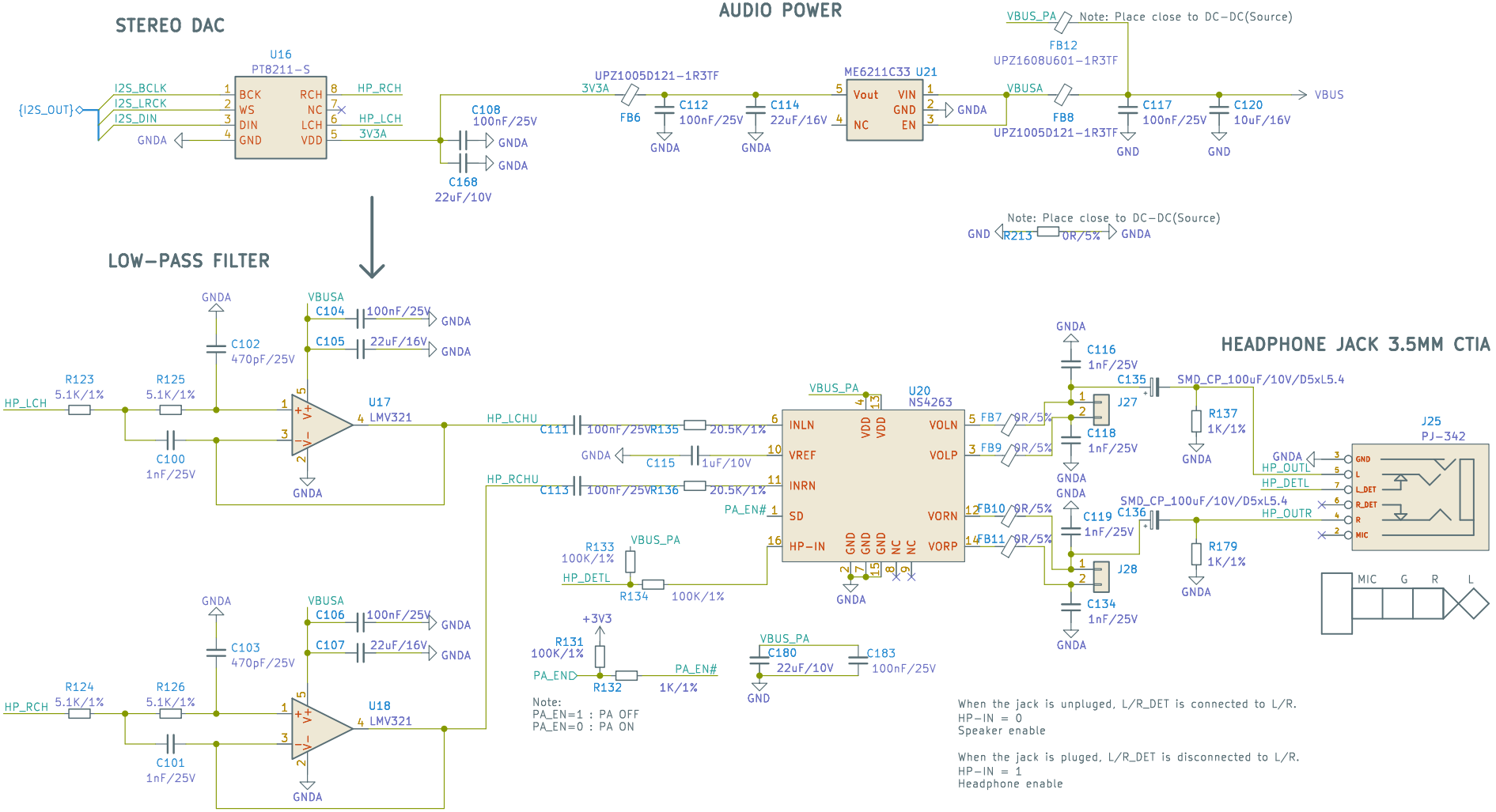
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Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
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FPGA-Audio



SiPEED

Sheet: /FPGA_AUDIO/		
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Title: Tang_Mega_138K_Dock		
Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 10/21

RGMII—ETHERNET

U10
RTL8211F-CG

PHY0_MDI0+	1	MDIP0	
PHY0_MDI0−	2	MDIN0	
PHY0_MDI1+	4	MDIP1	
PHY0_MDI1−	5	MDIN1	
PHY0_MDI2+	6	MDIP2	
PHY0_MDI2−	7	MDIN2	
PHY0_MDI3+	9	MDIP3	
PHY0_MDI3−	10	MDIN3	
PHY0_CLK	37	XTAL_IN	
	36	XTAL_OUT/EXT_CLK	
	35	CLKOUT	
PHY0_TXD0	18	TXD0	
PHY0_TXD1	17	TXD1	
PHY0_TXD2	16	TXD2	
PHY0_TXD3	15	TXD3	
PHY0_TXEN	19	TXCTL	
PHY0_TXCLK	20	TXCLK	
PHY0_RXD0/RXDLY	25	RXD0/RXDLY	
PHY0_RXD1/TXDLY	24	RXD1/TXDLY	
PHY0_RXD2/PLLOFF	23	RXD2/PLLOFF	
PHY0_RXD3/AD0	22	RXD3/PHYAD0	
PHY0_RXCLK/AD1	27	RXC/PHYAD1	
PHY0_RXDV/AD2	26	RXCTL/PHY_AD2	
	13	MDC	PHY0_MDCK, R25 1.5K/5%, R91 1.5K/5%
	14	MDIO	PHY0_MDIO
	12	PHYRSTB	PHY0_RSTN → PHY_VIO
	31	INTB/PMEB	R83 5.1K/1% → GND; C60 10nF/50V → GND
	32	LED0/CFG_EXT	PHY0_LED0/CFG_EXT
	33	LED1/CFG_LDO0	PHY0_LED1/CFG_LDO0
	34	LED2/CFG_LDO1	PHY0_LED2/CFG_LDO1
	39	RSET	R84 2.49K/1% → GND
	30	REG_OUT/LDO_OUT	PHY0_REG_SW, L4 CIGT201610LH2R2MNE, PHY0_REG_1V0; C61 4.7uF/10V, C63 100nF/25V
	29	DVDD33	PHY0_DVDD_3V3
	21	DVDD10	PHY0_DVDD_1V0
	28	DVDD_RG	PHY_VIO
	3, 8, 38	AVDD10_1, AVDD10_2, AVDD10_3	PHY0_AVDD_1V0
	11, 40	AVDD33_1, AVDD33_2	PHY0_AVDD_3V3
	41	GND	GND

R76 0R/5% (GND ↔ PHY0_CLK)
TP5 COPB

R24 0R/5% (GND ↔ GNDS)

+3V3 → FB3 UPZ1005D121-1R3TF → PHY0_AVDD_3V3 (PWR_FLAG)
C65 100nF/25V, C67 100nF/25V → GND

+3V3 → FB4 UPZ1005D121-1R3TF → PHY0_DVDD_3V3 (PWR_FLAG)
C69 4.7uF/10V, C71 100nF/25V → GND

PHY0_REG_1V0 → R95 0R/5% → PHY0_DVDD_1V0 (PWR_FLAG)
C64 100nF/25V → GND

PHY0_REG_1V0 → R96 0R/5% → PHY0_AVDD_1V0 (PWR_FLAG)
C68 100nF/25V, C70 100nF/25V, C72 100nF/25V → GND

TR1
H82409S

PHY0_MDI3−	R35 2.2R/1%	R_MDI3−	D12 UDD32C03LO1
PHY0_MDI3+	R97 2.2R/1%	R_MDI3+	
PHY0_MDI2−	R98 2.2R/1%	R_MDI2−	D13 UDD32C03LO1
PHY0_MDI2+	R100 2.2R/1%	R_MDI2+	
PHY0_MDI1−	R101 2.2R/1%	R_MDI1−	D14 UDD32C03LO1
PHY0_MDI1+	R102 2.2R/1%	R_MDI1+	
PHY0_MDI0−	R103 2.2R/1%	R_MDI0−	D15 UDD32C03LO1
PHY0_MDI0+	R107 2.2R/1%	R_MDI0+	

TR1 pins: 2, 1, 3, 5, 4, 6, 8, 7, 9, 11, 10, 12
TR1 → ETH0_MDI3−, ETH0_MDI3+, ETH0_MDI2−, ETH0_MDI2+, ETH0_MDI1−, ETH0_MDI1+, ETH0_MDI0−, ETH0_MDI0+
TR1 → R106 75R/1%, GD1 SMB90XM, C96 1nF/2KV → GNDS
C66 10nF/50V → GND

{RGMII}
Resistors should be placed close to SOC

RGMII_TXCK	R77	0R/5%	PHY0_TXCLK
RGMII_TXEN	R78	0R/5%	PHY0_TXEN
RGMII_TXD0	RN4 ORx4/5%		PHY0_TXD0
RGMII_TXD1	RN4 ORx4/5%		PHY0_TXD1
RGMII_TXD2	RN4 ORx4/5%		PHY0_TXD2
RGMII_TXD3	RN4 ORx4/5%		PHY0_TXD3
RGMII_MDIO	R87	0R/5%	PHY0_MDIO
RGMII_MDC	R222	0R/5%	PHY0_MDCK

{EPHY_CTRL}

{RGMII}
Resistors should be placed close to PHY

RGMII_RX0	RN6 ORx4/5%		PHY0_RXD0/RXDLY
RGMII_RX1	RN6 ORx4/5%		PHY0_RXD1/TXDLY
RGMII_RX2	RN6 ORx4/5%		PHY0_RXD2/PLLOFF
RGMII_RX3	RN6 ORx4/5%		PHY0_RXD3/AD0

{EPHY_CTRL}

EPHY_RST	R218	0R/5%	PHY0_RSTN
EPHY_CLK	R219	0R/5%	PHY0_CLK
RGMII_RXCK	R220	0R/5%	PHY0_RXCLK/AD1
RGMII_RXDV	R221	0R/5%	PHY0_RXDV/AD2

{RGMII}

+3V3 → FB2 UPZ1005D121-1R3TF → PHY_VIO (PWR_FLAG)
C57 100nF/25V, C58 4.7uF/10V, C59 100nF/25V, C62 100nF/25V → GND

PHY0 Config
PHY0 Address = 3'b001

PHY_VIO
PHY0_RXDV/AD2 R85 5.1K/1%
PHY0_RXCLK/AD1 R86 5.1K/1%
PHY0_RXD3/AD0 R79 5.1K/1%
PHY0_RXD2/PLLOFF R80 5.1K/1% R88 5.1K/1%
PHY0_RXD1/TXDLY R81 5.1K/1% R89 5.1K/1%
PHY0_RXD0/RXDLY R82 5.1K/1% R90 5.1K/1%
→ GND

PHY0 Address = 3'b001
Pull Down, PPL coontinue toggling@ALDPS Mode
Pull Up to add 2ns delay to TXC/RXC for TXD/RXD latching

RGMII IO Power	CFG_EXT	CFG_LDO[1:0]
External 3.3V	1'b1	2'b00
External 1.8V	1'b1	2'b10
Internal 1.8V	1'b0	2'b10

PHY_VIO R92 5.1K/1% PHY0_LED0/CFG_EXT
Pull up to use external power source for IO

PHY0_LED2/CFG_LDO1 R93 5.1K/1%
PHY0_LED1/CFG_LDO0 R94 5.1K/1% → GND
CFG_LDO[1:1]=0x11, select 3.3V external power

J12
HC-WK88-H865T-QB

ETH0_MDI3−	8	PHY0_LED0/CFG_EXT LED3	Orange R28 3.3K/1% → PHY_VIO
ETH0_MDI3+	7	PHY0_LED1/CFG_LDO0 LED5	Orange R29 3.3K/1% → GND
ETH0_MDI1−	6	PHY0_LED2/CFG_LDO1 LED11	Orange R30 3.3K/1% → GND
ETH0_MDI2−	5		
ETH0_MDI2+	4		
ETH0_MDI1+	3		
ETH0_MDI0−	2		
ETH0_MDI0+	1		
SH			

SH → GNDS

SiPEED

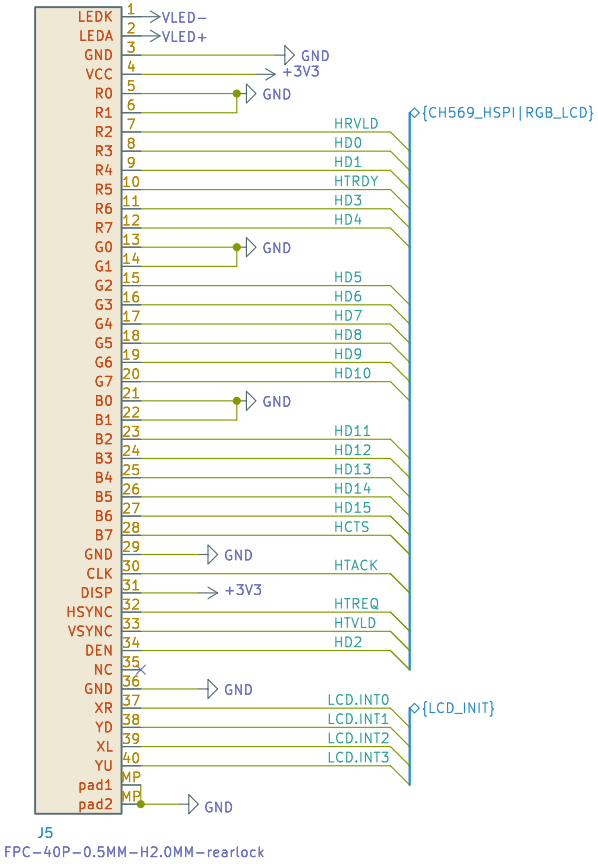
Sheet: /RGMII_ETHNET/
File: RGMII_sch.kicad_sch

Title: Tang_Mega_138K_Dock

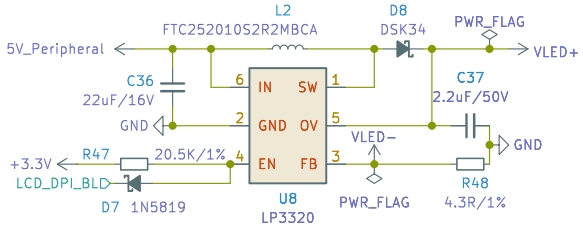
Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 11/21

RGB LCD Display

RGB LCD Display Connector

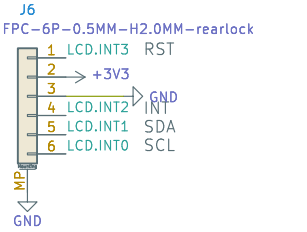


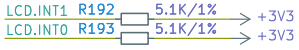
BACKLIGHT Driver



Feedback Voltage = 250mV
BackLight Current = 250mV/4.3R ≈ 58.1mA
7inch LCD QTM070TFT40-001 : IF=20*3=60mA

Capacitive Touch Connector





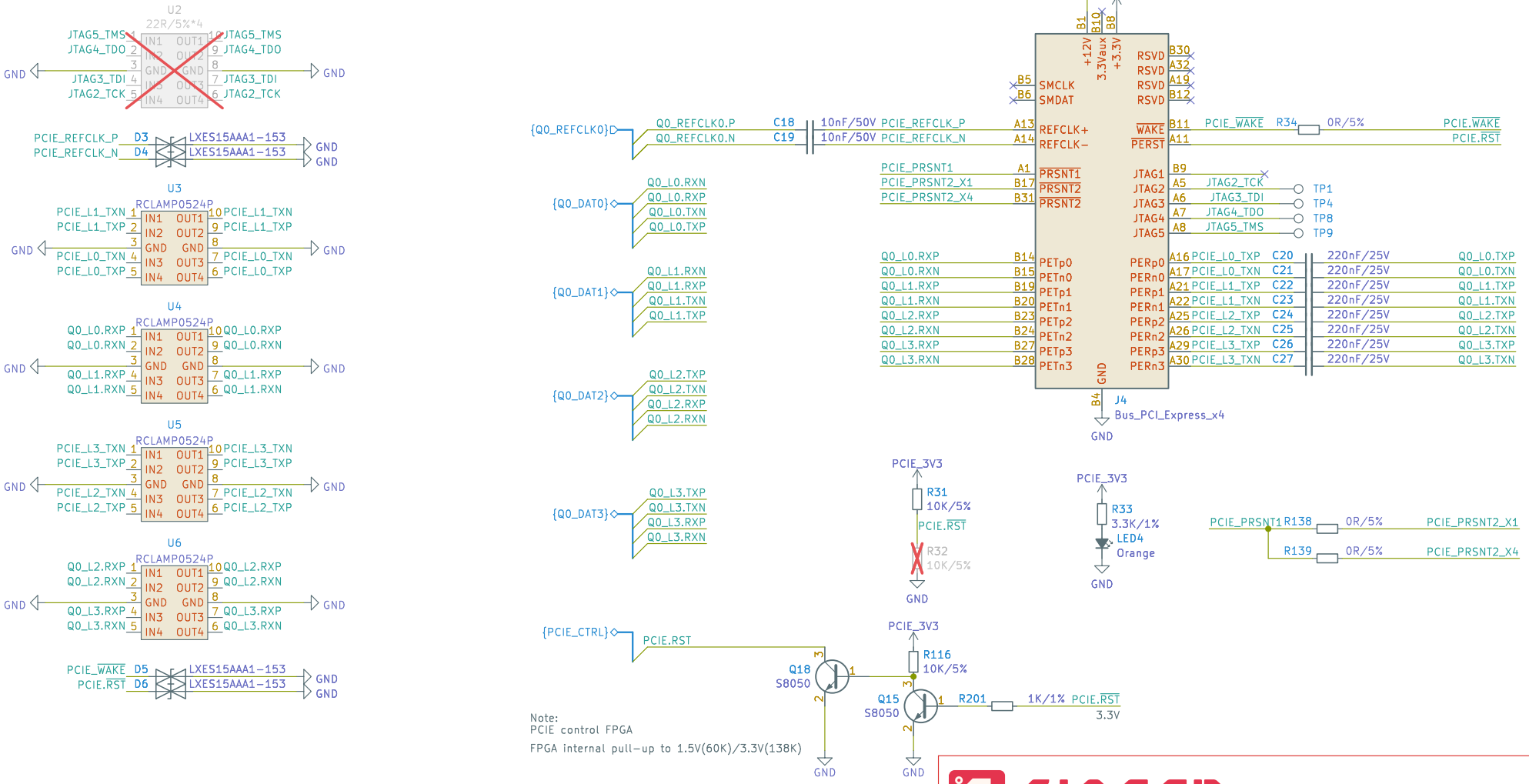
SiPEED

Sheet: /SYS_DPI/
File: LCD_DPI.kicad_sch

Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 12/21

PCI-Express X4

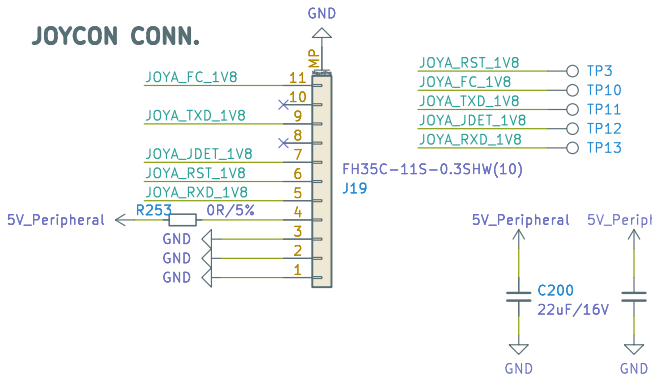


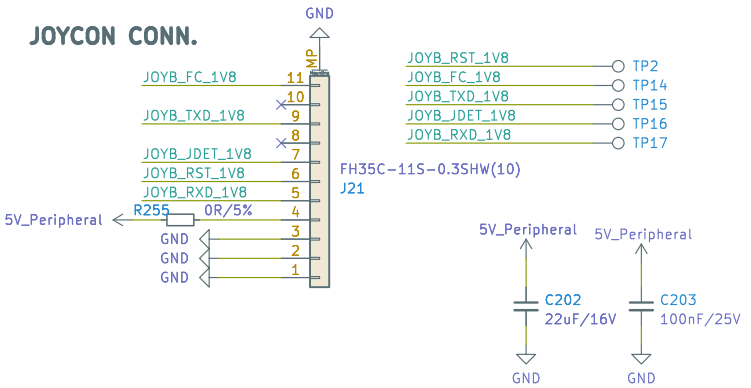
SiPEED

Sheet: /PCIE_GT/
File: SERDES_PCIE.kicad_sch

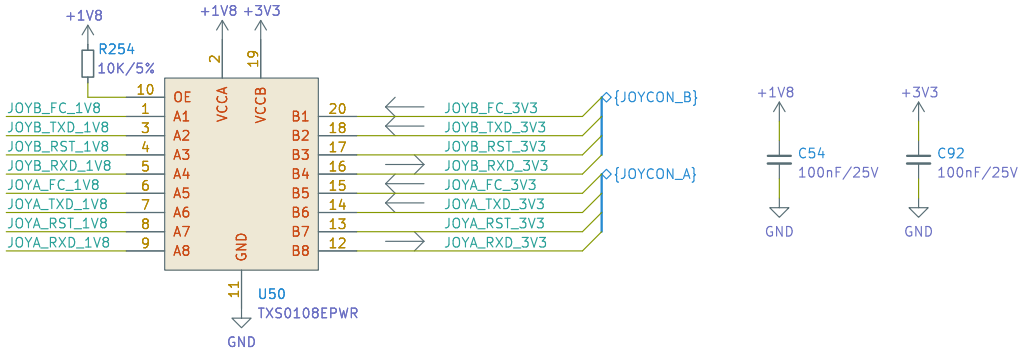
Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 13/21





Bi-directional Voltage Translator

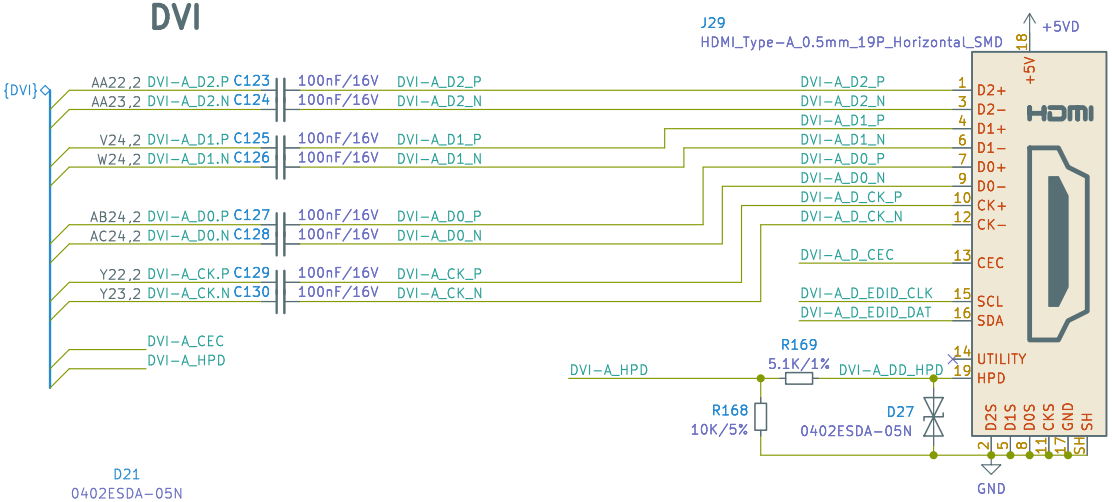


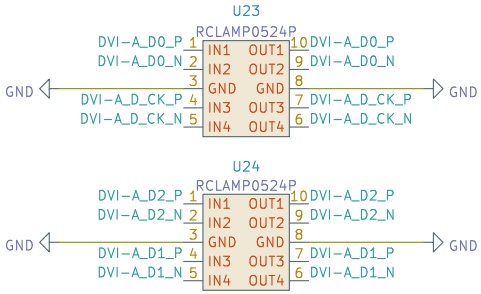
Note:
RST=0: Enable JOYCON
RST=1: Disable JOYCON

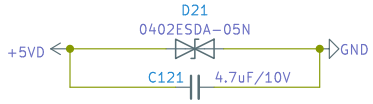
JDET=1:
HIGH when connected to console via Bluetooth.
Joy-Con will not send serial data post-handshake unless pin is pulled LOW by console.

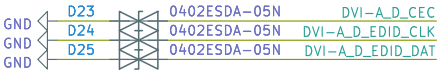
Sheet: /JOYCON/ File: JOYCON.kicad_sch		
Title: Tang_Mega_138K_Dock		
Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 14/21

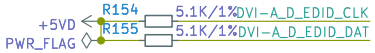
DVI

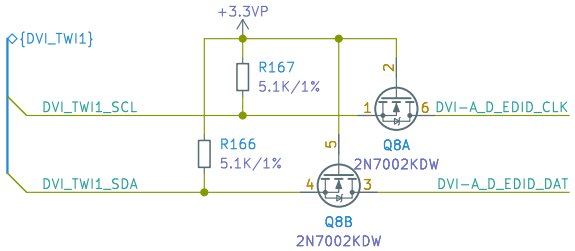








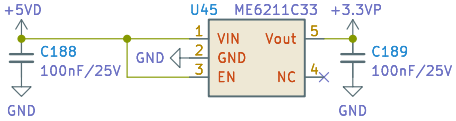


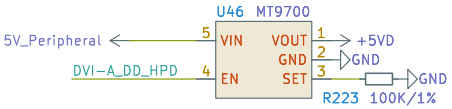




DVI Power control

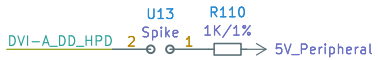
(Save Battery Power)

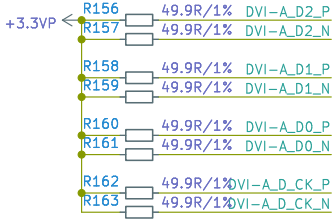


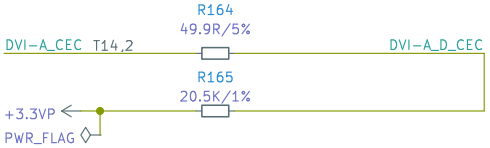


Insert HDMI : HPD=1
Unplug HDMI : HPD=0

Current limit: 68mA







SiPEED

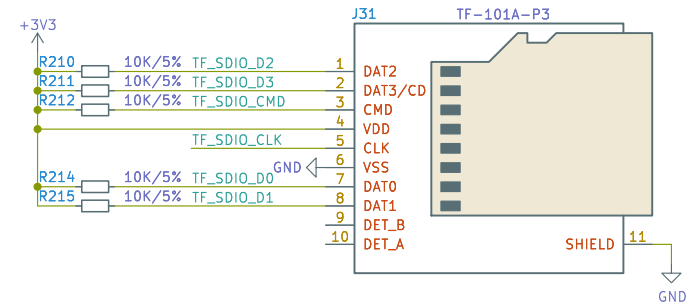
Sheet: /FPGA_TMDS_A/
File: DVI_A.kicad_sch

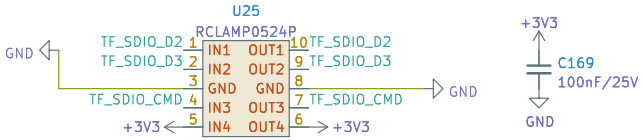
Title: Tang_Mega_138K_Dock

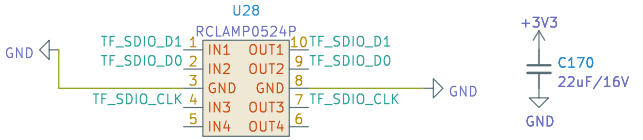
Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 15/21

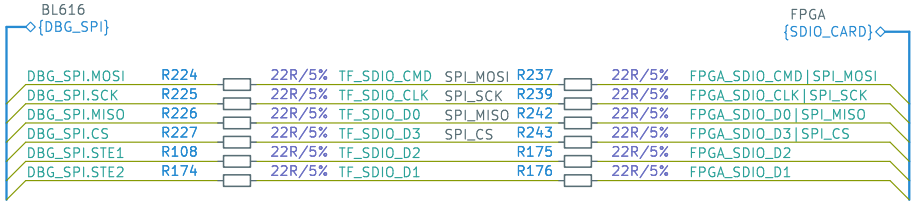
STORAGE

BL616 MICRO SD CARD





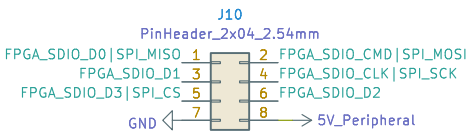




EVB mode : BL616_SPI high-Z , FPGA can work in SPI mode or SDIO mode

Game mode : BL616_SPI Master , FPGA_SPI Slave , FPGA_SPI_CS diasble , FPGA_SDIO_D1/D2 work as SPI_CS(input) or other function

Q1 : How to switch between EVB mode and Game mode?
A1: Switch DBG_ModeSel through SW2(6P Switch DIP)



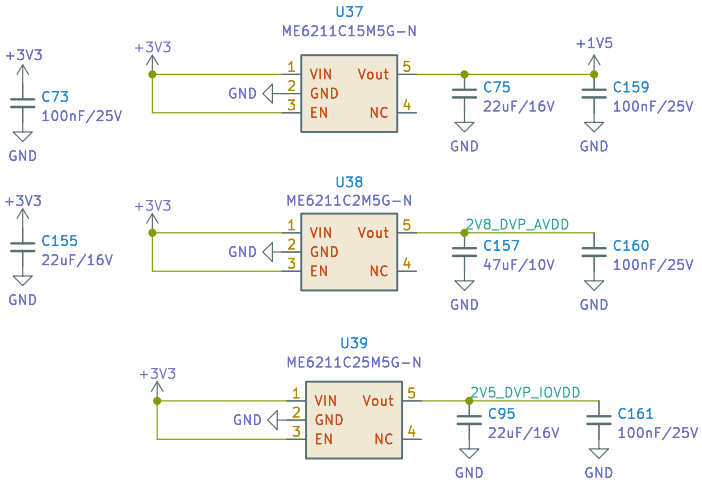
SiPEED

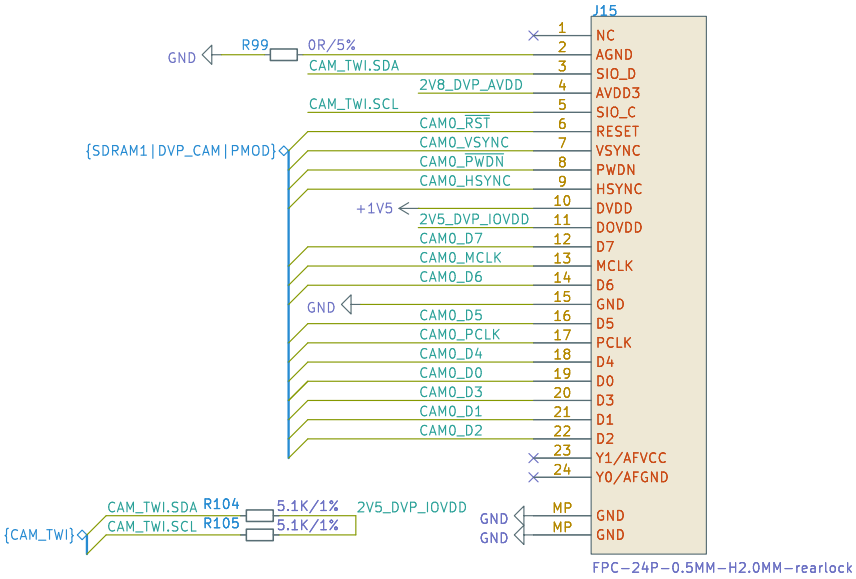
Sheet: /SYS_STORAGE/
File: SDIO&M.2.kicad_sch

Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 16/21

DVP-CAMERA





CAM0_D0 is IOB135A/CLKHOLD_N, should be pull-up in SSPI MODE

SiPEED

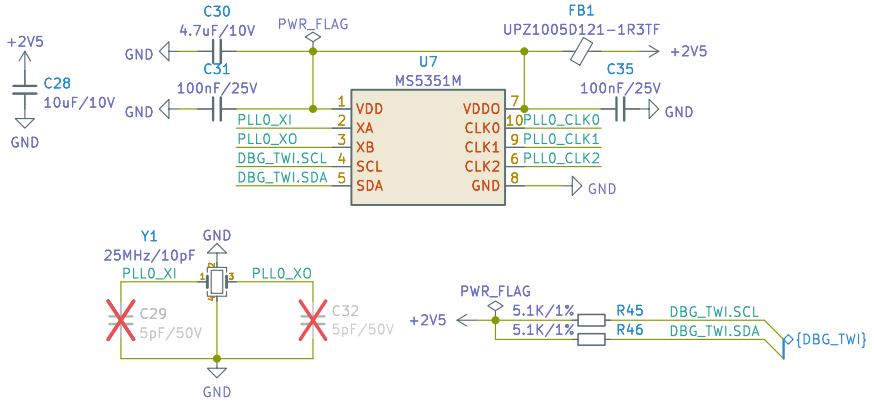
Sheet: /DVP_CAMERA/
File: DVP_CAM.kicad_sch

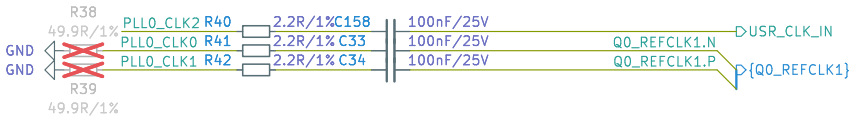
Title: Tang_Mega_138K_Dock

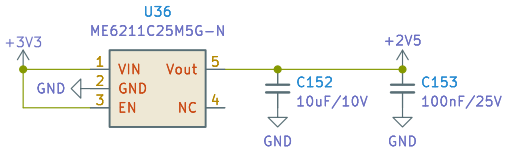
Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 17/21

CLOCK

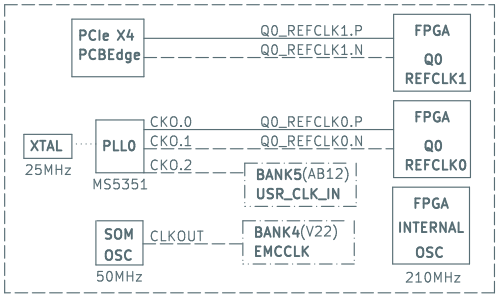
EXT. PLL







COLCK TREE



FPGA internal clock generator OSC.
The user can change the value of the option "Frequency Divider" to generate a different output clock frequency, which can be one of 64 values: 1/2, 1/3, 1/4, 1/6, 1/8, 1/10, ... or 1/126 of the oscillator frequency.

SiPEED

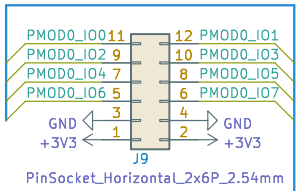
Sheet: /SYS_CLOCK/
File: CLOCK.kicad_sch

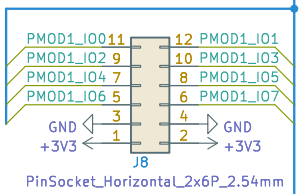
Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 18/21

EX CONN.

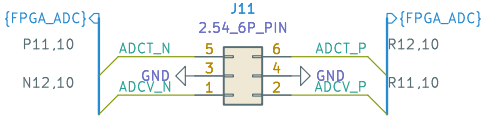
PMOD



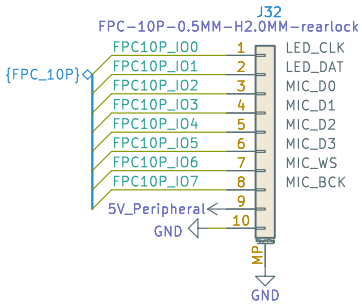


{SDRAM1 | DVP_CAM | PMOD}

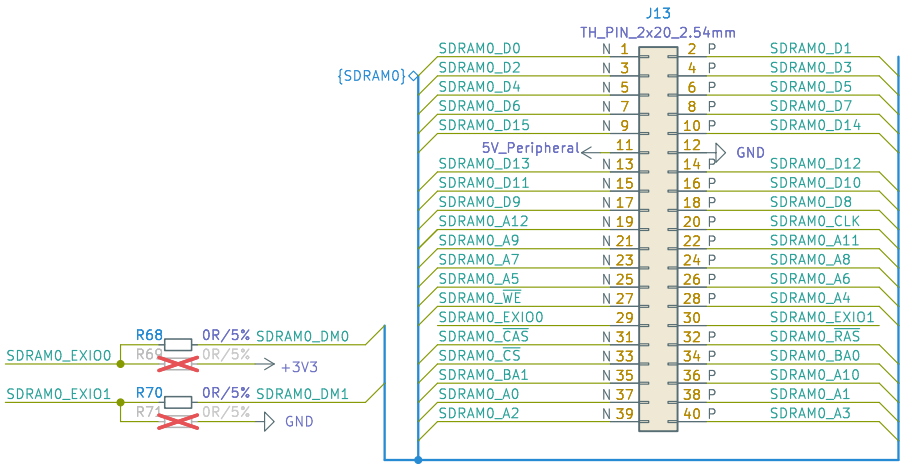
ADC CONN.



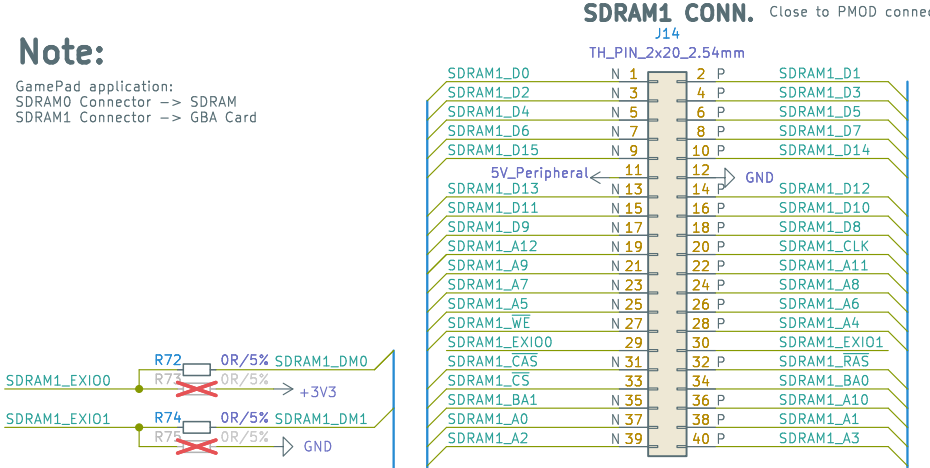
FPC10P CONN.



SDRAM0 CONN. Close to PCIE connector



SDRAM1 CONN. Close to PMOD connector



{SDRAM1 | DVP_CAM | PMOD}

Note:

GamePad application:
SDRAM0 Connector -> SDRAM
SDRAM1 Connector -> GBA Card

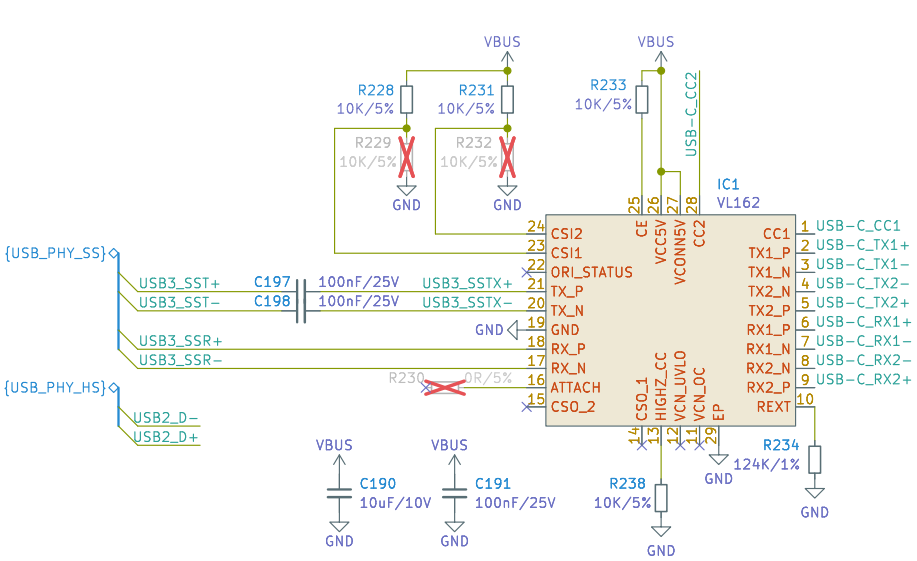
SiPEED

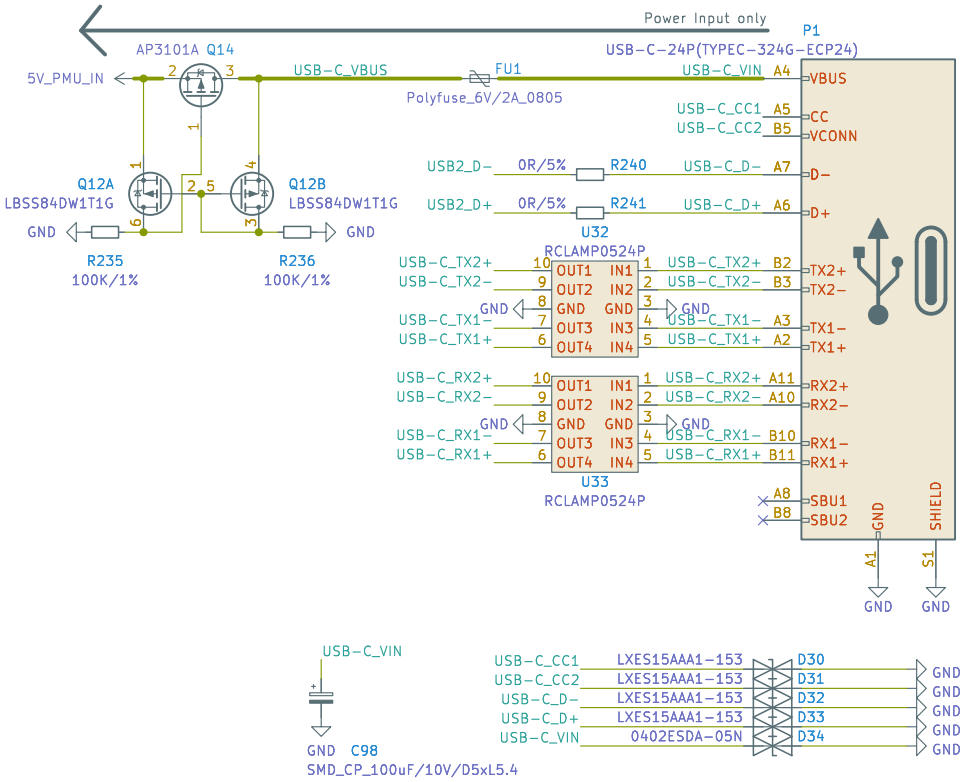
Sheet: /FPGA_EXT_CONN./
File: EX_CONN..kicad_sch

Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 19/21

CH569 ⟷ USB 2:1 Data Switch ⟷ CH569 USB-C Connector





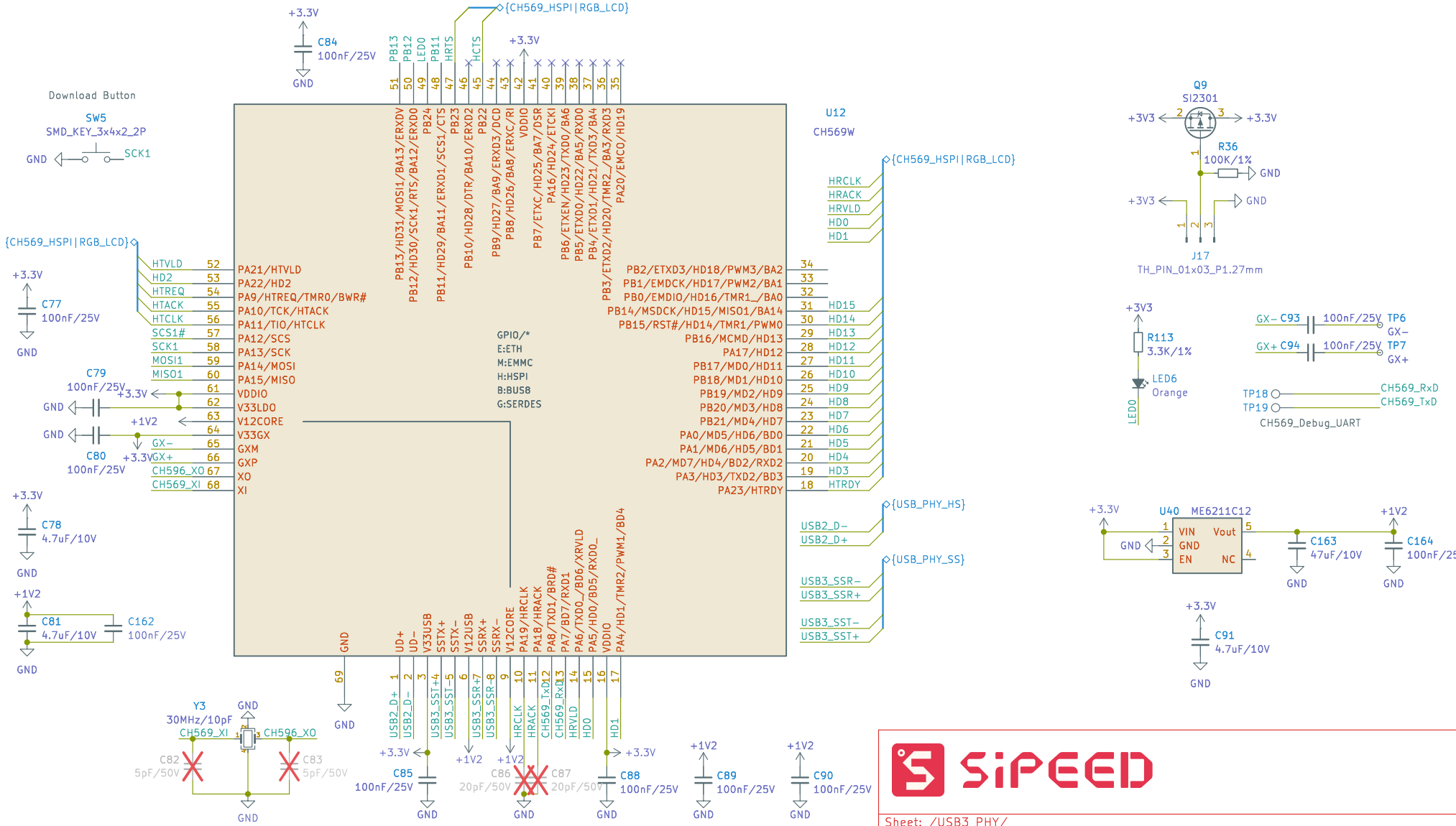
SiPEED

Sheet: /SYS_USB-C_DRP/
File: USBC_UFP.kicad_sch

Title: Tang_Mega_138K_Dock

Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 20/21

USB3-PHY



Sheet: /USB3_PHY/		
File: USB3_PHY.kicad_sch		
Title: Tang_Mega_138K_Dock		
Size: A4	Date: 2024-10-30	**Rev: 1.2**
KiCad E.D.A. 8.0.6		Id: 21/21